In [1]:

```
import pandas as pd
df = pd.read_csv(r'C:\Users\KHAN\Desktop\sp-500.csv')
print(df)
```

```
    Exchange Symbol                     Shortname                      Longname  \
0        NMS   AAPL                    Apple Inc.                    Apple Inc.   
1        NMS   NVDA            NVIDIA Corporation            NVIDIA Corporation   
2        NMS   MSFT         Microsoft Corporation         Microsoft Corporation   
3        NMS   AMZN              Amazon.com, Inc.              Amazon.com, Inc.   
4        NMS  GOOGL                 Alphabet Inc.                 Alphabet Inc.   
..       ...    ...                           ...                           ...   
497      NMS    CZR  Caesars Entertainment, Inc.  Caesars Entertainment, Inc.   
498      NYQ    BWA              BorgWarner Inc.              BorgWarner Inc.   
499      NMS   QRVO                  Qorvo, Inc.                  Qorvo, Inc.   
500      NYQ    FMC               FMC Corporation               FMC Corporation   
501      NYQ   AMTM       Amentum Holdings, Inc.       Amentum Holdings, Inc.   

                     Sector                        Industry  Currentprice  \
0                Technology            Consumer Electronics        254.49   
1                Technology                  Semiconductors        134.70   
2                Technology       Software - Infrastructure        436.60   
3         Consumer Cyclical                 Internet Retail        224.92   
4    Communication Services  Internet Content & Information        191.41   
..                      ...                             ...           ...   
497       Consumer Cyclical               Resorts & Casinos         32.82   
498       Consumer Cyclical                      Auto Parts         31.88   
499              Technology                  Semiconductors         70.85   
500         Basic Materials              Agricultural Inputs         50.15   
501             Industrials     Specialty Business Services         19.17   

         Marketcap        Ebitda  Revenuegrowth           City State  \
0    3846819807232  1.346610e+11          0.061      Cupertino    CA   
1    3298803056640  6.118400e+10          1.224    Santa Clara    CA   
2    3246068596736  1.365520e+11          0.160        Redmond    WA   
3    2365033807872  1.115830e+11          0.110        Seattle    WA   
4    2351625142272  1.234700e+11          0.151  Mountain View    CA   
..             ...           ...            ...            ...   ...   
497     6973593600  3.668000e+09         -0.040           Reno    NV   
498     6972155904  1.882000e+09         -0.048   Auburn Hills    MI   
499     6697217024  6.731300e+08         -0.052     Greensboro    NC   
500     6260525568  7.033000e+08          0.085   Philadelphia    PA   
501     4664099328  4.330000e+08         -0.031      Chantilly    VA   

           Country  Fulltimeemployees  \
0    United States           164000.0   
1    United States            29600.0   
```

```
2    United States          228000.0
3    United States         1551000.0
4    United States          181269.0
..             ...               ...
497  United States           51000.0
498  United States           39900.0
499  United States            8700.0
500  United States            5800.0
501  United States               NaN

                                  Longbusinesssummary    Weight
0    Apple Inc. designs, manufactures, and markets ...  0.069209
1    NVIDIA Corporation provides graphics and compu...  0.059350
2    Microsoft Corporation develops and supports so...  0.058401
3    Amazon.com, Inc. engages in the retail sale of...  0.042550
4    Alphabet Inc. offers various products and plat...  0.042309
..                                                 ...       ...
497  Caesars Entertainment, Inc. operates as a gami...  0.000125
498  BorgWarner Inc., together with its subsidiarie...  0.000125
499  Qorvo, Inc. engages in development and commerc...  0.000120
500  FMC Corporation, an agricultural sciences comp...  0.000113
501  Amentum Holdings, Inc. provides engineering an...  0.000084

[502 rows x 16 columns]
```

In [2]:
```
df = df[['Exchange', 'Sector', 'Industry', 'Currentprice', 'Marketcap', \
         'Ebitda', 'Revenuegrowth', 'Country', 'Fulltimeemployees', 'State']]
```

In [3]:
```
print(df)
```

```
    Exchange                 Sector                        Industry  \
0        NMS             Technology            Consumer Electronics
1        NMS             Technology                  Semiconductors
2        NMS             Technology      Software - Infrastructure
3        NMS      Consumer Cyclical                 Internet Retail
4        NMS  Communication Services  Internet Content & Information
..       ...                    ...                             ...
497      NMS      Consumer Cyclical              Resorts & Casinos
498      NYQ      Consumer Cyclical                     Auto Parts
499      NMS             Technology                  Semiconductors
500      NYQ        Basic Materials            Agricultural Inputs
501      NYQ            Industrials     Specialty Business Services

     Currentprice      Marketcap        Ebitda  Revenuegrowth        Country  \
0          254.49  3846819807232  1.346610e+11          0.061  United States
1          134.70  3298803056640  6.118400e+10          1.224  United States
2          436.60  3246068596736  1.365520e+11          0.160  United States
3          224.92  2365033807872  1.115830e+11          0.110  United States
4          191.41  2351625142272  1.234700e+11          0.151  United States
..            ...            ...           ...            ...            ...
497         32.82     6973593600  3.668000e+09         -0.040  United States
498         31.88     6972155904  1.882000e+09         -0.048  United States
499         70.85     6697217024  6.731300e+08         -0.052  United States
500         50.15     6260525568  7.033000e+08          0.085  United States
501         19.17     4664099328  4.330000e+08         -0.031  United States

     Fulltimeemployees State
0             164000.0    CA
1              29600.0    CA
2             228000.0    WA
3            1551000.0    WA
4             181269.0    CA
..                 ...   ...
497            51000.0    NV
498            39900.0    MI
499             8700.0    NC
500             5800.0    PA
501                NaN    VA

[502 rows x 10 columns]
```

In [4]:

```
print(df.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 502 entries, 0 to 501
Data columns (total 10 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   Exchange           502 non-null    object
 1   Sector             502 non-null    object
 2   Industry           502 non-null    object
 3   Currentprice       502 non-null    float64
 4   Marketcap          502 non-null    int64
 5   Ebitda             473 non-null    float64
 6   Revenuegrowth      499 non-null    float64
 7   Country            502 non-null    object
 8   Fulltimeemployees  493 non-null    float64
 9   State              482 non-null    object
dtypes: float64(4), int64(1), object(5)
memory usage: 39.3+ KB
None
```

In [5]:
```python
print(df.duplicated().sum())
```

```
0
```

In [6]:
```python
df = df.dropna()
```

In [7]:
```python
print(df.isnull().sum())
```

```
Exchange             0
Sector               0
Industry             0
Currentprice         0
Marketcap            0
Ebitda               0
Revenuegrowth        0
Country              0
Fulltimeemployees    0
State                0
dtype: int64
```

In [8]:
```python
print(df['Exchange'].unique())
```

```
['NMS' 'NYQ' 'BTS' 'NGM']
```

In [9]:
```
print(df['Sector'].unique())
```

```
['Technology' 'Consumer Cyclical' 'Communication Services'
 'Financial Services' 'Consumer Defensive' 'Healthcare' 'Energy'
 'Industrials' 'Utilities' 'Real Estate' 'Basic Materials']
```

In [11]:
```
print(df['Industry'].unique())
```

```
['Consumer Electronics' 'Semiconductors' 'Software - Infrastructure'
 'Internet Retail' 'Internet Content & Information' 'Auto Manufacturers'
 'Insurance - Diversified' 'Discount Stores'
 'Drug Manufacturers - General' 'Credit Services' 'Oil & Gas Integrated'
 'Healthcare Plans' 'Household & Personal Products'
 'Home Improvement Retail' 'Entertainment' 'Software - Application'
 'Beverages - Non-Alcoholic' 'Telecom Services' 'Communication Equipment'
 'Restaurants' 'Information Technology Services' 'Diagnostics & Research'
 'Medical Devices' 'Tobacco' 'Medical Instruments & Supplies'
 'Aerospace & Defense' 'Farm & Heavy Construction Machinery'
 'Travel Services' 'Asset Management' 'Financial Data & Stock Exchanges'
 'Conglomerates' 'Utilities - Regulated Electric' 'Computer Hardware'
 'Insurance - Property & Casualty' 'Railroads' 'Apparel Retail'
 'Semiconductor Equipment & Materials' 'Oil & Gas E&P'
 'Footwear & Accessories' 'Integrated Freight & Logistics'
 'Insurance Brokers' 'Biotechnology' 'REIT - Industrial'
 'Utilities - Renewable' 'REIT - Specialty' 'Specialty Chemicals'
 'Electronic Components' 'Specialty Industrial Machinery'
 'Waste Management' 'Confectioners' 'Lodging'
 'REIT - Healthcare Facilities' 'Medical Care Facilities'
 'Specialty Business Services' 'Drug Manufacturers - Specialty & Generic'
 'Medical Distribution' 'Specialty Retail' 'Oil & Gas Midstream'
 'REIT - Retail' 'Building Products & Equipment' 'Insurance - Life'
 'Copper' 'Utilities - Diversified' 'Industrial Distribution'
 'Oil & Gas Equipment & Services' 'Engineering & Construction'
 'Utilities - Independent Power Producers' 'Rental & Leasing Services'
 'Oil & Gas Refining & Marketing' 'Grocery Stores' 'Gold'
 'Beverages - Brewers' 'Real Estate Services' 'Agricultural Inputs'
 'Airlines' 'Consulting Services' 'Electronic Gaming & Multimedia'
 'Trucking' 'Resorts & Casinos' 'Food Distribution'
 'Residential Construction' 'Packaged Foods' 'Health Information Services'
 'Building Materials' 'REIT - Residential' 'REIT - Diversified'
 'Scientific & Technical Instruments' 'Chemicals' 'Steel'
 'Pollution & Treatment Controls' 'Utilities - Regulated Water'
 'Farm Products' 'Electrical Equipment & Parts' 'Personal Services'
 'Utilities - Regulated Gas' 'Packaging & Containers' 'Solar'
 'Beverages - Wineries & Distilleries' 'Tools & Accessories'
 'Advertising Agencies' 'REIT - Office' 'Auto Parts' 'Luxury Goods'
 'Apparel Manufacturing' 'Auto & Truck Dealerships' 'REIT - Hotel & Motel'
 'Capital Markets' 'Pharmaceutical Retailers' 'Leisure'
 'Furnishings, Fixtures & Appliances']
```

In [8]:

```
import numpy as np
df['Industry'] = np.where(df['Industry'].str.contains\
    ('Semiconductor Equipment & Materials|Software - Infrastructure|Software - Application\
    |Internet Retail|Internet Content & Information|Communication Equipment|Consumer Electronics\
    |Telecom Services|Information Technology Services|Electronic Components|Computer Hardware\
    |Electronic Gaming & Multimedia|Scientific & Technical Instruments\
    |Electrical Equipment & Parts', na=False), 'Semiconductors', df['Industry'])

df['Industry'] = np.where(df['Industry'].str.contains('Drug Manufacturers - Specialty & Generic\
   |Biotechnology|Medical Devices|Chemicals|Pharmaceutical Retailers|Medical Instruments & Supplies\
   |Diagnostics & Research|Drug Manufacturers - General|Medical Care Facilities|Medical Distribution\
   |Healthcare Plans|Health Information Services|REIT - Healthcare Facilities', na=False)\
                       , 'Medical field', df['Industry'])

df['Industry'] = np.where(df['Industry'].str.contains('Insurance - Property & Casualty|Insurance - Life\
   |Insurance Brokers|Credit Services|Asset Management|Capital Markets|Financial Data & Stock Exchanges\
   |Conglomerates|Insurance - Diversified|Travel Services|Specialty Business Services|REIT - Industrial\
   |REIT - Specialty|REIT - Retail|REIT - Residential|REIT - Diversified|REIT - Office|REIT - Hotel & Motel\
   |Real Estate Services|Airlines|Consulting Services|Building Materials|Rental & Leasing Services\
   |Financial & Hospitality', na=False), 'Business Sector', df['Industry'])

df['Industry'] = np.where(df['Industry'].str.contains('Home Improvement Retail|Household & Personal Products\
   |Restaurants|Apparel Retail|Footwear & Accessories|Specialty Retail|Grocery Stores|Confectioners\
   |Packaged Foods|Discount Stores|Beverages - Non-Alcoholic|Beverages - Brewers\
   |Beverages - Wineries & Distilleries|Tobacco|Luxury Goods|Apparel Manufacturing|Auto Parts\
   |Auto Manufacturers|Auto & Truck Dealerships|Furnishings, Fixtures & Appliances|Personal Services\
   |Advertising Agencies|Tools & Accessories|Leisure|Food Distribution|Entertainment|Lodging|Resort', na=False),
    'Consumer Goods', df['Industry'])

df['Industry'] = np.where(df['Industry'].str.contains('Oil & Gas E&P|Oil & Gas Midstream\
   |Oil & Gas Refining & Marketing|Oil & Gas Equipment & Services|Utilities - Regulated Electric\
   |Utilities - Renewable|Oil & Gas Integrated|Utilities - Diversified|Utilities - Independent Power Producers\
   |Utilities - Regulated Water|Utilities - Regulated Gas|Solar|Copper|Gold|Steel\
   |Farm & Heavy Construction Machinery|Aerospace & Defense|Railroads|Integrated Freight & Logistics\
   |Trucking|Engineering & Construction|Industrial Distribution|Residential Construction\
   |Specialty Chemicals|Specialty Industrial Machinery|Waste Management|Pollution & Treatment Controls\
   |Agricultural Inputs|Farm Products|Packaging & Containers\
   |Building Products & Equipment', na=False), 'Industerial & utilities', df['Industry'])
```

In [9]:

```
print(df['Industry'].unique())
```

```
['Semiconductors' 'Consumer Goods' 'Business Sector' 'Medical field'
 'Industerial & utilities']
```

In [10]:
```
print(df['Country'].unique())
```

```
['United States' 'Canada']
```

In [13]:
```
print(df['Sector'].unique())
```

```
['Technology' 'Consumer Cyclical' 'Communication Services'
 'Financial Services' 'Consumer Defensive' 'Healthcare' 'Energy'
 'Industrials' 'Utilities' 'Real Estate' 'Basic Materials']
```

In [14]:
```
import os
desktop = os.path.join(os.path.expanduser('~'), 'Desktop')
df.to_csv(r'C:\Users\KHAN\Desktop\sp-500(2).csv')
```